Journal of Science and Medicine in Sport xxx (xxxx)

Contents lists available at ScienceDirect

Journal of Science and Medicine in Sport

journal homepage: www.elsevier.com/locate/jsams

Review

Are return-to-play times longer in lower-limb muscle injuries involving the intramuscular tendon? A systematic review

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ARTICLE INFO

Article history:
Received 20 March 2023
Received in revised form 14 September 2023
Accepted 2 October 2023
Available online xxxx

Keywords:
Elite athletes
Muscle injury
Tendon tear
Rehabilitation
BAMIC
Prognosis

ABSTRACT

Objectives: To determine the influence of severity and direction (craniocaudal length *vs* cross-sectional area) of intramuscular tendon tears in the lower limb on return-to-play times, compared to muscle injuries without intramuscular tendon involvement.
Design: Systematic review with meta-analysis.
Methods: AMED, CINAHL, SPORTDiscus, ScienceDirect, PubMed (MEDLINE) and Web-of-Science were searched from inception to 31st July 2023, retrieving 666 records, of which nine were deemed eligible. A random-effects meta-analysis was performed on time to return-to-play for British Athletics Muscle Injury Classification 'b' *vs* 'c'.
Results: On the Quality in Prognosis Studies tool, one study had low risk of bias and eight had high risk. Using a best-evidence synthesis, no strong evidence emerged for a difference in time to return-to-play between injuries with or without intramuscular tendon involvement. Moderate evidence was found for an association between increased return-to-play time and the presence of "waviness" on magnetic resonance imaging and loss of tendon tension, but no association with longitudinal extent of tendon involvement. Pooled analysis revealed a medium effect-size difference between British Athletics Muscle Injury Classification 'b' and 'c' injuries, favouring classification 'b' (Hedges $g = 0.67$; 95% confidence interval 0.20 to 1.15; $P = 0.002$).
Conclusions: It remains difficult to provide an accurate prognosis for muscle injuries involving the intramuscular tendon due to high risk of bias and moderate heterogeneity across studies. Moderate evidence favoured the prognosis for injuries at the musculotendinous junction (British Athletics Muscle Injury Classification 'b') over intratendinous injuries (British Athletics Muscle Injury Classification 'c').
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Practical implications

- Determining the prognosis of lower-limb muscle injuries with IMT involvement remains difficult due to the lack of strong evidence available.
- There is moderate evidence that favours the prognosis of BAMIC 'b' injuries, with fewer days to RTP than for intratendinous 'c' injuries.
- Clinicians should remain cautious during the rehabilitation of muscle injuries involving the IMT that have a large percentage cross-sectional area and presence of waviness and/or loss of tendon tension on MRI.

1. Introduction

Non-contact muscle injuries of the lower limb are a substantial problem for professional athletes and their clubs.[1] Muscle injury is one of the most frequent causes of absence from training and competition, being the third most common sports-related injury at the 2016 Summer Olympic Games.[2] Absence can have significant economic and competitive implications for the athlete and all stakeholders involved,[3,4] placing considerable pressure on medical staff for an accurate prognosis on time to return to play (RTP).

There has been increasing interest in the literature in the prognosis of muscle injuries involving the intramuscular tendon (IMT).[5] The IMT (central tendon or aponeurosis) is a section of the tendon that extends along and into the muscle and acts as a central supporting strut to which the muscle fibres attach.[6] It has been suggested that injuries with IMT involvement require prolonged rehabilitation.[7,8] This extended rehabilitation has been attributed to anatomical differences in fibre orientation, collagen

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https://doi.org/10.1016/j.jsams.2023.10.002
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content and vascularity of tendinous tissue in comparison to muscle tissue.[9] Early research from a series of 62 injuries in Australian rules football and rugby players, confirmed by magnetic resonance imaging (MRI), reported a median RTP of 72 days in biceps femoris injuries with IMT disruption. In comparison, those injuries with no IMT involvement took a median of 21 days to RTP.[10] Similarly, Pollock et al., who reviewed 65 hamstring injuries in 44 track-and-field athletes, found that injuries involving the IMT were associated with increased RTP times (mean 27.0, SD 6.8 days), compared to injuries that were isolated to the muscle belly (mean 21.0, SD 10.2 days).[11] Prolonged rehabilitation times have further been documented in the quadriceps and soleus/gastrocnemius muscles with IMT involvement.[12,13]

Recently, there has been dispute regarding the relevance of IMT involvement, as other studies have found minimal or no significant difference in RTP times in comparison with injuries not involving the IMT.[5,14] McAuley et al. observed hamstring injuries over four years in professional footballers, reporting no difference in British Athletics Muscle Injury Classification (BAMIC) grade two injuries with and without IMT involvement.[14] The BAMIC scale (Appendix 1, Table S1) was proposed by Pollock et al. to specify the grade (0–4) and site ('a' = myofascial, 'b' = musculotendinous junction, or 'c' = intratendinous) of muscle injury.[15] It appears to be a reliable classification system[16] that has previously been employed in respect of time to RTP.[11]

The literature is equivocal regarding muscle injury prognosis with IMT involvement. The orientation of tears (craniocaudal length/longitudinal *vs* cross-sectional area), as well as the presence of "waviness" (loss of tendon tension causing an irregular contour) on MRI may have significant implications for RTP,[5] yet these prognostic features are not always addressed within research.[14] The methodological quality of studies has also been questioned due to bias from a lack of practitioner blinding to MRI results.[5] For the reasons outlined above, accurate prognosis in terms of RTP between injuries either with or without IMT involvement is challenging for practitioners. The aim of this systematic review was to determine the influence of severity and orientation of IMT tears in the lower limb on RTP times, compared to injuries within the muscle belly.

2. Methods

2.1. Reporting standard

This systematic review conforms to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) guidelines.[17] The study was registered in the PROSPERO prospective register of systematic reviews prior to study initiation (CRD42022343009).

2.2. Literature search strategy and information sources

A systematic, computerized English-language literature search of electronic databases – AMED (1985–present), CINAHL (1981–present), SPORTDiscus (1949–present), ScienceDirect (1997–present), PubMed (MEDLINE, 1946–present) and Web-of-Science (1997–present) – was conducted and ended on 31st July 2023. An overview of the electronic search syntax using Boolean operators in titles, abstracts, and keywords of indexed documents can be found in Appendix 2 (Table S2).

Secondary searches were undertaken on the reference lists of all selected articles, which were screened manually to retrieve articles potentially missed during the computerized search. In addition, forward reference searching using Web-of-Science was undertaken to explore potential follow-up studies through citations. Automation tools were applied in electronic databases, where possible, for article type to exclude review articles (systematic review, narrative-review, meta-analyses). All references were exported into bibliographic software (EndNote Web) and identified duplicates removed. Author CEB independently carried out the searches for study selection.

2.3. Screening for eligibility

Two independent reviewers (CEB and SAP) manually screened the titles and abstracts of the remaining publications for eligibility in accordance with the inclusion/exclusion criteria (Table 1). In instances where the title and abstract contained insufficient detail to indicate whether an article was potentially relevant to the review, the complete article was obtained and read. Disagreements between the reviewers were resolved by consensus.

2.4. Study selection

A study selection tool was used to help identify studies that met the eligibility criteria (Appendix 2, Table S3). In instances where the study did not examine the relationship between IMT injury and RTP in elite athletes, the paper was excluded from the review.

2.5. Data extraction

Data extraction was undertaken by CEB and an independent reviewer (SAP) validated the extracted data once extraction was completed. The following data were extracted from the included studies: 1) study authors and date; 2) number of participants, their age and the level of sport they compete/perform at (*e.g.*, professional players from a Premier League division football team in England); 3) mode of investigation used to determine muscle/tendon injury (*e.g.* MRI 1.5 T/3 T); 4) muscle grading system used to determine the extent of muscle

Table 1
Inclusion and exclusion criteria.

Inclusion criteria	
Population	Healthy adult participants (18+ years of age) with acute lower-limb muscle injury involving the IMT.
Elite sport	Only studies conducted in elite sport. For example, Association football/soccer, Rugby football (*e.g.*, Rugby Union or Rugby League), Gridiron football (*e.g.*, American football or Canadian football), Australian rules football, Gaelic football and Track and Field.
Study design	Randomized controlled trial, cohort study, case–control study, case series, or prospective/retrospective observational study design.
Diagnosis of injury	Confirmed diagnosis by use of magnetic resonance imaging (1.5 T or 3.0 T).
Muscle grading	Reporting of muscle injuries using a muscle grading system or equivalent (*e.g.*, British Athletic Muscle Injury Classification or Peetrons). If a muscle grading system was not reported, then the MRI features of length and cross-sectional area (CSA) of tendon/muscle disruption should be reported for a grading system to be applied.
Outcomes	Reporting of days missed through injury or return-to-play time frames.
Comparison	Only studies that compare outcome measures between muscle injuries with and without IMT involvement.
Publication	Peer-reviewed publication in the English language.
Exclusion criteria	
Review articles	Systematic reviews, meta-analyses, narrative reviews
Grey literature	Conference abstracts or other 'grey' literature, including unpublished studies and theses.

and/or tendon disruption (*e.g.* BAMIC and Peetrons); 5) muscle group affected (*e.g.* hamstrings, quadriceps, soleus/gastrocnemius); 6) definition of RTP; 7) days taken for RTP; 8) additional findings on RTP time for grade and orientation of muscle/tendon injury. If the data on RTP time (days) could not be extracted from the published studies, an e-mail including a list of the data of interest was sent to the corresponding author of the study. Data were considered missing if no replies had been received from the authors two weeks after a second email.

2.6. Quality assessment

Quality assessment of the articles was conducted by two independent reviewers (CEB and SAP) using a modified version of the Quality in Prognosis Studies (QUIPS) tool described by Hayden et al.[18] (Appendix 3, Table S5). The QUIPS tool has been recommended by the Cochrane Prognosis Methods Group and has been used in comparable reviews assessing the prognosis of muscle injuries.[19,20] QUIPS assesses six potential bias domains, each consisting of specific items of opportunity for bias. The criteria were modified and agreed upon using the prompting items and considerations presented by Reurink et al.[21] so as to reflect our research question.

As described by Hayden et al., 75 % or more of the QUIPS items had to be fulfilled for a domain to be considered low in bias.[18] The overall risk of bias for each study was calculated to be low if five out of the six areas were scored as low-risk, including domain four relating to outcome measurement of time to RTP.[21] If there was a difference in opinion on an item, a consensus was reached by the two reviewers (CEB and SAP). If no consensus was reached, the independent opinion of a third reviewer (JW) was decisive.

2.7. Data synthesis

To assist with synthesizing the information contained within the reports and identifying potential differences in time to RTP (days) between muscular (BAMIC 'b') and IMT (BAMIC 'c') injuries, unadjusted mean differences and 95 % confidence intervals (CIs) were calculated for those graded using the BAMIC scale. If the mean and standard deviation (SD) were not reported, the median and interquartile range were used to impute these values using established methods described by Shi et al.[22] and Luo et al.,[23] respectively.

Pooled summary statistics were calculated from studies reporting the BAMIC scale using a random effects meta-analysis, with a DerSimonian–Laird estimator, to take account of heterogeneity across studies. This was conducted using the Meta-Essentials workbook for differences between independent groups.[24] Grouped data (means, SDs, and sample sizes) were used to calculate Hedges' *g* standardized effect size,[25] interpreted as small (0.20), medium (0.5) and large (0.8). In addition to a 95 % CI for the pooled estimate, a 95 % prediction interval was calculated. This represents the boundaries within which 95 % of the estimates of other similar studies would be expected to lie; in the presence of heterogeneity this interval will be wider than the corresponding 95 % CI.[26] The *I*-squared (I^2) statistic was used to quantify study heterogeneity, expressed as a percentage from 0 % to 100 %. This indicates the percentage of observed variance that reflects real differences in effect size rather than sampling error, with an I^2 value <25 % indicative of low heterogeneity, a value from 26 % to 74 % indicative of moderate heterogeneity, and a value >75 % indicative of high heterogeneity.[27] Lastly, a best-evidence synthesis was used, consisting of five levels of evidence-based qualitative assessment[28]: 1) strong evidence: provided by two or more studies with low risk of bias and by generally consistent findings in all studies (75 % of the studies reported consistent findings); 2) moderate evidence: provided by one study with low risk of bias and/or two or more studies with high risk of bias and by generally consistent findings in all studies (75 % of the studies reported consistent findings); 3) limited evidence: provided by only one study with high risk of bias; 4) conflicting evidence: inconsistent findings in multiple studies (<75 % of the studies reported consistent findings); and 5) no evidence: when no studies could be found. These criteria for best-evidence synthesis have been used in recent systematic reviews of risk factors and prognosis in muscle injury.[29–31]

3. Results

3.1. Search results

The primary search revealed 666 articles identified from databases and a further three articles *via* reference lists and citation searching. Once automation tools were applied and duplicates were removed, 241 manuscripts obtained *via* databases remained in the reference manager (EndNote Web). After examining titles and abstracts, 15 were sought for retrieval and retained for full-text analysis. When assessing for eligibility, a total of eight were deemed eligible and seven excluded. A further three reports identified *via* reference lists or citation searching were assessed for eligibility, with an additional one article deemed eligible. Nine reports were thereby included in the review. A flowchart of the literature search[17] and the number of records identified through the study selection process can be found in Fig. 1. Reasons for excluding articles are given in Table S4, Appendix 2.

3.2. Study characteristics

The participant characteristics for each study included within the review are shown in Table 2. In total, 441 injuries from a pool of 319 athletes (57 female) were included across the nine studies. The mean (SD) age of participants was 24.7 (3.9) years, representing athletic populations from cohorts across football,[8,14,32] Australian Football League,[7,12] and track and field.[11,33,34] Seven studies reported hamstring injuries and two reported quadriceps injuries.

3.3. Diagnosis of muscle injury

Among the eligible studies, two different grading scales were used (6 studies used BAMIC and one study used both BAMIC and Peetrons) to diagnose the severity of injury and type of tissue involvement. Two studies reported on the MRI features without applying a muscle grading classification.[5,12] Four studies used an MRI Tesla strength of 1.5 T[5,8,12,32] and one study used 3 T[34]; in the remaining studies each participant was scanned with either 1.5 T or 3 T MRI.[7,11,14,33]

3.4. Methodological quality

The scores on the potential risk of bias domains for all eligible studies are shown in Table 3. The details of the methodological quality assessment are outlined in Appendix 3, Table S6. One study[5] scored a low risk of bias, meeting five out of the six domains, whilst the remaining eight studies scored a high risk of bias. For the specific items within the bias domains there was disagreement on 23 out of the 270 items assessed (8.5 %), on which consensus was subsequently reached by the two reviewers. A high rate of attrition and incomplete, inconsistent or poor-quality prognostic factor measurement were the most common areas of bias across studies: 9 of 9 (100 %) and 7 of 9 (78 %) at high risk of bias, respectively. The other sources of bias related to confounding variables (3 of 9 [33 %] studies at high risk of bias) and outcome measurement (1 of 9 [11 %] studies at high risk of bias).

3.5. Data synthesis

The best evidence synthesis is summarized in Table 4.

3.5.1. Severity (grade)

There was conflicting evidence for a difference in time to RTP in BAMIC grade one, two and three injuries, irrespective of the

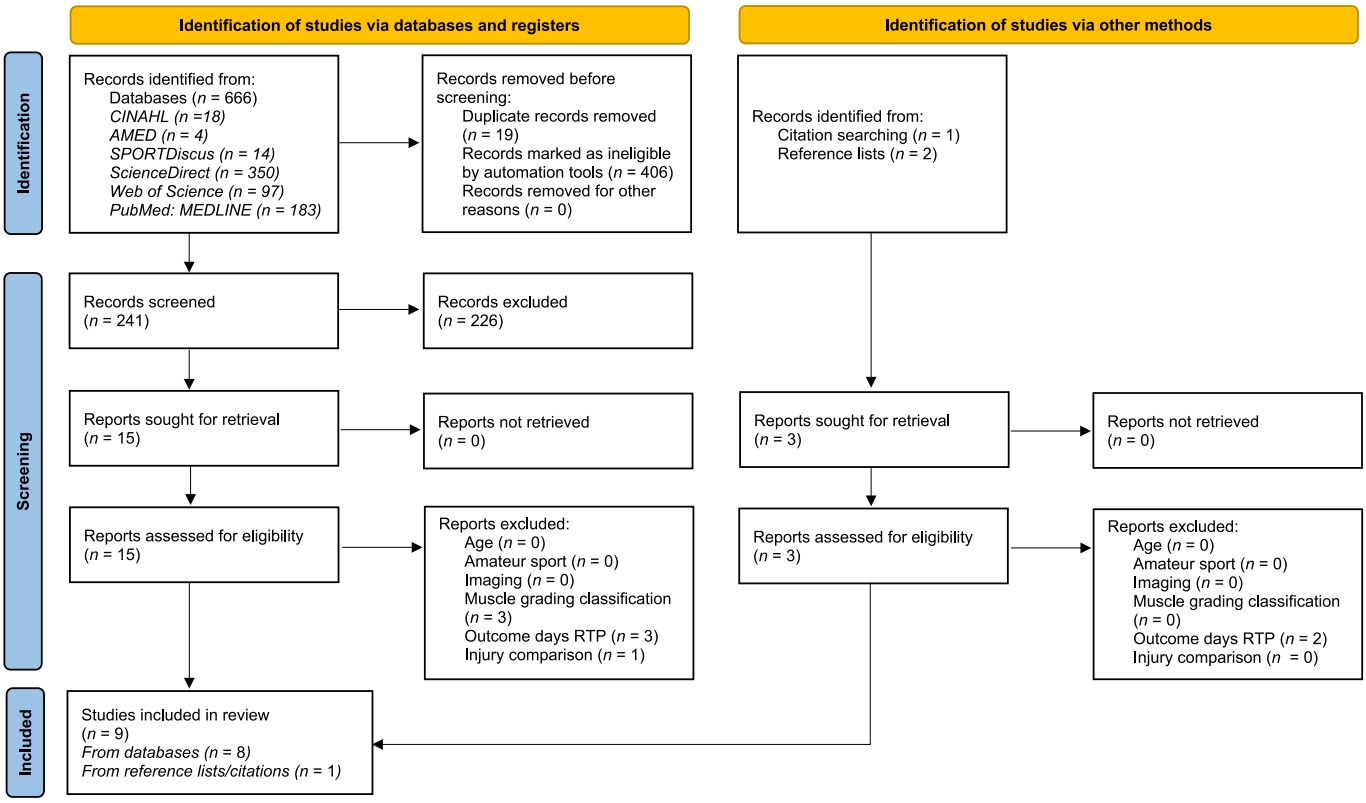

Fig. 1. PRISMA 2020 flow diagram[17] of the study selection process.

subclassification for type of tissue involved.[8,11,14,32–34] Two studies reported significant differences in time to RTP between grade one and two and one and three injures, but no difference in time to RTP between grade two and three injuries.[33,34] Two studies with a high risk of bias provided moderate evidence for a positive association on correlation analysis between severity of muscle grading and increased time to RTP.[11,32]

3.5.2. Location

Moderate evidence, based on two studies with high risk of bias, demonstrated that quadriceps injuries with IMT involvement carry a less favourable prognosis than injuries without IMT involvement.[12,33] McAleer et al. reported a significant difference in RTP time for BAMIC class 'c' injuries ($P = 0.001$) relative to the other classes.[33] Cross et al. did not report their findings using the BAMIC classification, but once length and cross-sectional area (CSA) of injuries were converted their findings favoured injuries without IMT involvement (Table 2).[12] The remaining seven studies investigated hamstring injuries, of which six directly examined the difference in RTP time between muscle injuries with and without IMT disruption. There was moderate evidence, based on five studies with high risk of bias and one study with low risk of bias, for increased time to RTP in those injuries with IMT disruption (Table 4).

3.5.3. BAMIC 'b' vs 'c' muscle injuries

The seven studies reporting RTP using the BAMIC scale are presented in Table 5. There was a favourable direction of effect regarding the BAMIC '2b' classification, with fewer days to RTP than for '2c' in five out of seven studies (moderate evidence).[8,11,32–34] Three out of four studies comparing BAMIC grade '3b' and '3c' injuries favoured the '3b' classification for fewer days to RTP (moderate evidence).[7,11,32,33]

The pooled analysis (Fig. 2) for BAMIC grade 2 and 3 injuries revealed a statistically significant difference in time to RTP between muscle injuries with IMT involvement ('2c' and '3c') and those without ('2b' and '3b'), favouring the BAMIC 'b' group: 11 estimates from seven studies (160 muscle injuries). The effect size was 0.67 (95 % CI 0.20, 1.15); $P = 0.002$. This corresponds to a mean difference of 8.93 days (95 % CI 1.63, 16.22) in favour of the BAMIC 'b' group. None of the CIs of included studies lay wholly outside the CI of the pooled estimate, indicating an absence of outlier effects. The 95 % prediction interval was −0.39, 1.73; 95 % of similar studies would therefore be expected to provide estimates within this range. The I^2 index of heterogeneity associated with the pooled estimate was moderate at 36.6 %, and the pooled effect should be interpreted accordingly. All seven studies included in the pooled analysis were rated as being of high risk of bias.

The effect size for the estimates relating specifically to '2b' *versus* '2c' injuries was 0.69 (95 % CI 0.13, 1.24; $I^2 = 49.7$ %) and for estimates relating specifically to '3b' *versus* '3c' injuries was similar at 0.64 (95 % CI −0.13, 1.40; $I^2 = 22.3$ %); the latter is, however, very imprecise owing to sparse data.

3.5.4. Craniocaudal length and cross-sectional area

There was moderate evidence of no association between longitudinal extent of tendon disruption and time to RTP from one study with low risk of bias and one study with high risk of bias.[5,34] Based on three studies, the percentage CSA of tendon disruption showed moderate evidence for an association with increased time to RTP.[32,34]

3.5.5. Waviness and loss of tendon tension

There was moderate evidence for the presence of waviness on MRI and loss of tendon tension being associated with increased time to RTP.[5,34] One study, with low risk of bias, reported mean (SD) time to RTP for injuries with and without waviness of 30.2 (10.8) and 22.6 (7.5), respectively.[5] However, there was no statistically significant difference when comparing the presence and absence of waviness in muscle injuries without IMT involvement.[5] The same study provided moderate evidence for significantly longer RTP times in those injuries with full-thickness IMT disruption (100 % of CSA) compared to injuries without IMT disruption.[5]

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Table 2
Details of participants, study characteristics and main findings on time to RTP.

Authors and date	Participant details	Number of injuries	Investigation	Muscle grading system	Muscle group	RTP definition	Days to RTP [no. of injuries]	Main findings	Risk of bias[a]
McAleer et al. (2022)[33]	Twenty-seven (male = 10, female = 17) Elite British Track and Field Athletes, mean (SD) age 24.7 (2.3) years.	38	MRI (1.5 T or 3 T)	BAMIC	Quadriceps	When an athlete completed maximal intensity sprinting without exacerbation.	Mean (SD): 0 = 9.3 (1.0) [6] 1a = 13.1 (5.8) [10] 1b = 12.0 (0.0) [1] 2a = 25.9 (9.9) [2] 2b = 23.5 (6.4) [3] 2c = 41.7 (18.7) [4] 3a = 26.0 (7.1) [2] 3b = 34.0 (0.0) [2] 3c = 34.8 (20.6) [4] 4c = 129.8 (16.8) [4]	In the BAMIC classification system (a–c), there was a significant difference in RTP time for BAMIC class c injuries ($P = 0.001$) relative to the other classes.	High
McAuley et al. (2022)[14]	Twenty-four male professional football players, mean (SD) age 26.0 (4.0) years.	32	MRI (1.5 T or 3 T)	BAMIC	Hamstring	The number of days until the player had resumed full team training and available for match selection.	Mean (SD): 0 = 8.0 (1.4) [2] 1a = 10.3 (7.3) [6] 1b = 14.2 (5.3) [5] 2a = 24.0 (10.4) [3] 2b = 26.0 (7.5) [6] 2c = 26.1 (8.2) [8] 3b = 29.0 (2.8) [2]	There were no differences in RTP between injuries with and without tendon involvement (1a *vs* 1b, or 2b *vs* 2c; both $P > 0.05$).	High
Pollock et al. (2022)[34]	Forty-six (male = 22, female = 24) Elite British Track and Field Athletes, mean (SD) age 24.6 (3.7) years.	70	MRI (3 T)	BAMIC	Hamstring	Completing unrestricted sprint efforts at full pace in spikes.	Median (IQR, range): 0 = 9.0 (4.3; 4–7) [16] 1a = 12.0 (4.0; 9–16) [6] 1b = 17.0 (10.3; 8–32) [19] 2a = 19.0 (3.5; 14–21) [4] 2b = 19.0 (3.5; 11–36) [11] 2c = 35.0 (9.5; 25–43) [7] 3a = 17.0 (0.0) [1] 3c = 51.5 (23.8; 28–70) [6]	There was a significant difference in RTP time between BAMIC class c *versus* class b ($P < 0.001$) injuries. Length of tendon injury was not associated with time to RTP ($P = 0.16$), but CSA was significantly associated with RTP time ($P = 0.03$). Loss of tendon tension was associated with a longer RTP time ($P = 0.02$).	High
Tears et al. (2022)[32]	Thirty male professional football players, mean (SD) age 26.26 (4.6) years.	39	MRI (1.5 T)	BAMIC Modified Peetrons	Hamstring	When they returned to full participation in team training and were available for match selection.	Mean (SD): Peetrons Grade 0 = 5.5 (0.5) [2] Grade 1 = 15.3 (12.2) [18] Grade 2 = 18.9 (12.3) [18] Grade 3 = 15.0 (0.0) [1] BAMIC 0a = 6.0 (0.0) [1] 0b = 13.0 (0.0) [1] 1a = 15.5 (14.9) [4] 1b = 7.0 (2.2) [3] 2a = 17.7 (8.3) [3] 2b = 14.9 (10.3) [18] 2c = 23.8 (5.8) [4] 3b = 25.3 (21.0) [3] 3c = 23.5 (11.5) [2]	A weak but statistically significant correlation between BAMIC and RTP ($r = 0.32$; 95 % CI 0.01 to 0.58; $P = 0.05$). Percentage CSA also demonstrated a weak and non-statistically significant correlation with RTP ($r = 0.02$; 95 % CI −0.30 to 0.33; $P = 0.91$).	High

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Table 2 (*continued*)

Authors and date	Participant details	Number of injuries	Investigation	Muscle grading system	Muscle group	RTP definition	Days to RTP [no. of injuries]	Main findings	Risk of bias[a]
Shamji et al. (2021)[8]	Thirty-six male professional football players, median age 26 years (IQR 6, range 18–36).	61	MRI (1.5 T)	BAMIC	Hamstring	No definition provided.	Median (IQR, range):[c] 0a = 8.00 (13.75, 5–22) [4] 0b = 13.00 (13.75, 7–2) [4] 1a = 8.00 (9.00, 2–14) [6] 1b = 17.00 (12.00, 3–34) [15] 2a = 12.00 (n/a, 3–25) [3] 2b = 22.00 (14.5, 6–39) [16] 2c = 37.00 (43.00, 8–103) [10] 3c = 15.00 (n/a, 9–36) [3]	Statistically significant difference in RTP time for anatomical site BAMIC (a–c) excluding grade 0 (P = 0.002) Mean rank RTP time of 24 days for no tendon involvement and 36 days for tendon involvement.	High
Eggleston et al. (2020)[7]	Twenty-four male professional Australian Rules players, mean (SD) age 23.5 (4.0) years.	41	MRI (1.5 T or 3 T)	BAMIC Four-grade (MR0, MR1, MR2, and MR3) injury classification based on MRI	Hamstring	Calculated from date of injury to date of the first game played post-injury.	Mean (SD):[c] 0 = 12.3 (2.9) [3] 1a = 20.0 (0.0) [1] 1b = 28.0 (0.0) [1] 2a = 24.5 (9.0) [4] 2b = 24.3 (4.9) [7] 2c = 21.4 (5.0) [5] 3a = 35 (0.0) [1] 3b = 19.5 (6.3) [8] 3c = 52.8 (30.1) [10] 4 = 35.0 (0.0) [1]	Hamstring injuries with high-grade IMT disruption on MRI were associated with increased time to RTP compared to injuries with low-grade or no IMT disruption. The longest RTP times were observed in injuries that had high-grade IMT disruption (longitudinal IMT disruption of > 5 cm in length or > 50 % of IMT CSA) involving the proximal biceps femoris IMT with concomitant injury of biceps femoris + semitendinosus muscles.	High
van der Made et al. (2018)[5]	Seventy professional male athletes from mixed sports, median age 24 (IQR 21–30).	70	MRI (1.5 T)	Own muscle grading system	Hamstring	The number of days from injury to successful completion of rehabilitation programme when sport-specific activities were performed unrestricted and pain free.	Mean (SD): No IMT disruption = 22.2 (7.0) [41] IMT disruption = 27.7 (10.0) [29] No waviness = 22.6 (7.5) [53] Waviness = 30.2 (10.8) [17] 2c = 24.0 (9.7) [5][b] 3c = 25.3 (8.6) [12][b] 4c = 30.2 (10.8) [17][b]	Only injuries with full-thickness IMT disruption were correlated with a significantly longer time to RTP compared with injuries without IMT disruption. There was no significant correlation between longitudinal tendon disruption and time to RTP.	Low

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Pollock et al. (2016)[11]	Forty-four (male = 28, female = 16) Elite British Track and Field Athletes, mean (SD) age 23.8 (4.3) years.	65	MRI (1.5 T or 3 T)	BAMIC	Hamstring	Until able to complete unrestricted training sprint efforts at full pace in spikes.	Mean (SD; range): 0 = 10.0 (4.7; 4–21) [21] 1a = 18.0 (3.8; 13–21) [5] 1b = 18.0 (11.0; 12–31) [4] 2a = 25.0 (5.7; 21–29) [2] 2b = 21.0 (10.2; 12–49) [17] 2c = 27.0 (6.8; 18–35) [8] 3b = 41.0 (0.0) [1] 3c = 84.0 (49.4; 40–128) [7]	Grade 3 severity ($P < 0.001$) and intratendinous (c) site ($P = 0.008$) were associated with an increase in RTP time. No difference in RTP times between hamstring muscle involved (biceps femoris long head *vs* semitendinosus *vs* semimembranosus *vs* biceps femoris short head).	High
Cross et al. (2004)[12]	Eighteen professional male Australian Rules Football players, mean (SD) age 23.7 (4.1) years.	25	MRI (1.5 T)	MRI features	Quadriceps	The number of days from the injury until the player returned to full team training.	Mean (SE): Central tendon (CT) = 26.85 (1.75) [7] Peripheral (P) = 9.17 (1.70) [8] 2b = 6.8 (0.10)[b] 3c = 30.0 (3.38)[b] RF-CT length 1–7 cm = 14.00 (0.00) [1] RF-CT length > 13 = 32.66 (6.02) [6] RF-CT CSA 1–14 % = 14.00 (4.34) [1] RF-CT CSA 15–24 % = 30.00 (3.51) [3] RF-CT CSA >25 % = 35.33 (2.51) [3] RF-P length 1–7 cm = 9.75 (3.77) [4] RF-P length 8–12 cm = 7.00 (2.83) [2] RF-P length > 13 cm = 8.50 (4.95) [2] RF-P CSA 1–14 % = 0 (2.51) [3] RF-P CSA 15–24 % = 9.67 (2.51) [3] RF-P CSA >25 % = 8.50 (3.07) [2]	Prognosis is significantly dependent on both the site and the size (CSA% and the length independently predictive) of the muscle strain injury. RF-CT diagnosis carries the least favourable prognosis ($P = 0.001$) MRI category RF-CT and CSA 15–24 % and >25 % indicate statistically significant different interactions from other MRI categories (peripheral) with the same CSA% for rehabilitation time.	High

SD = standard deviation, SE = standard error, IQR = interquartile range, MRI = magnetic resonance imaging, BAMIC = British Athletic Muscle Injury Classification, RTP = return to play, CSA = cross-sectional area. T = Tesla magnetic strength, IMA = intramuscular aponeurosis, IMT = intramuscular tendon, CT = central tendon, RF-CT = rectus femoris central tendon, RF-P = rectus femoris peripheral.

[a] See Table 3 for the specific domains of risk of bias for each study.

[b] Converted to BAMIC grading based scale on CSA and/or longitudinal extent of injury.

[c] Injury recurrences or exacerbations were included in the analysis of RTP.

Table 3
Risk of bias.

Date	Study author	Potential risk of bias domains[a]						Risk of bias[b]
		1	2	3	4	5	6	
2022	McAleer et al.[33]	+	−	−	+	+	+	High
2022	McAuley et al.[14]	+	−	−	+	+	+	High
2022	Pollock et al.[34]	+	−	−	+	+	+	High
2022	Tears et al.[32]	+	−	+	+	−	+	High
2021	Shamji et al.[8]	+	−	−	−	+	+	High
2020	Eggleston et al.[7]	+	−	−	+	−	+	High
2018	van der Made et al.[5]	+	−	+	+	+	+	Low
2016	Pollock et al.[11]	+	−	−	+	+	+	High
2004	Cross et al.[12]	+	−	−	+	−	+	High

+ = potential risk of bias limited sufficiently, − = potential risk of bias.
[a] Domain 1: study participation, domain 2: study attrition, domain 3: prognostic factor measurement, domain 4: outcome measurement, domain 5: confounding measurement and account, domain 6: analysis.
[b] Low risk of bias requires positive scores for a minimum five out of six items, to include item 4.

4. Discussion

The aim of this systematic review was to determine the influence of severity and orientation of IMT tears in the lower limb on RTP times, compared to muscle injuries without IMT involvement, with a specific focus on lower-limb injuries in BAMIC categories '2b', '2c', '3b' and '3c'. A robust search of the literature was conducted with analysis and synthesis of information from nine studies relating to prognosis of muscle injuries involving the IMT. The main finding is that there is no strong evidence for a difference in time to RTP between injuries with and without IMT involvement. Nonetheless, the pooled analysis for grade two and three injuries revealed a medium effect-size difference between BAMIC 'b' and 'c' muscle injuries, with a favourable direction of effect towards the BAMIC '2b' classification of fewer days to RTP than for '2c'. However, involvement of the IMT is not the sole factor for increased time to RTP and several prognostic features – such as the percentage CSA of tendon disruption, presence of waviness and loss of tendon tension on MRI – all showed moderate evidence for increased time to RTP. The longitudinal extent of tendon disruption showed no association and further research is required to determine the influence of orientation of fibre disruption on time to RTP.

To the authors' knowledge, this is the first review to directly compare RTP timeframes in muscle injuries with and without IMT involvement. Previous reviews have explored prognostic indicators on MRI for predicting time to RTP in hamstring injuries.[21,35] Reurink et al. concluded that there is no strong evidence to support an association between any MRI variables and time to RTP, similarly reporting high risk of bias in all but one included study.[21] Fournier-Farley et al. reviewed clinical and MRI features of hamstring injuries that influenced RTP and concluded that IMT involvement was one of the identified factors for a longer recovery.[35] Many of the studies included were graded as level IV evidence according to the Oxford Centre for Evidence-Based Medicine. In agreement with our findings, the poor methodological quality of included studies may contribute to the contradictory findings across these systematic reviews.

4.1. Methodological limitations in the studies

There is a lack of high-quality studies examining the prognosis of muscle injuries with IMT involvement. Heterogeneity across studies should be acknowledged, and sample sizes ranged from 18[12] to 70,[5] representing elite athletes from diverse sporting backgrounds. The definition used for RTP was variable between studies and only two of the included studies reported that the clinicians involved in the rehabilitation were blinded to the MRI findings.[5,11] Research has previously advocated a protective rehabilitation period for IMT injuries,[11] causing a potential area for clinician bias in delaying initial loading during rehabilitation, leading to longer RTP times.[34] Further, the measurement of prognostic factors was variable within and between studies, with images acquired by either 1.5 T or 3.0 T MRI. Greater diagnostic accuracy may be achieved by using 3.0 T MRI, due to the superior imaging quality.[36] There was some inconsistency regarding the inclusion of injury recurrences or exacerbations in the calculation of RTP; most studies excluded recurrences, but two included them.[7,8] The included studies were predominantly retrospective cohort studies (Table S4, Appendix 3); the nature of these studies allows research to be conducted within an elite sport setting, where there is less control over confounding variables.

4.2. Anatomy and physiological healing of tendon tissue

Knowledge of the anatomy and physiological healing of tendon, in comparison to muscle, is important in selecting the appropriate nature and duration of rehabilitation. Tendons are the functional link for transferring muscle contraction force to the skeletal system to produce movement.[37] Tendons have higher collagen content (70 % tendon dry weight) than muscle (1–10 % of muscle mass dry weight) and a parallel fibre orientation to provide tensile strength for their important role in muscle–tendon unit force production.[37,38] Research has documented that tendon tissue heals differently from muscle as it is less vascularized and requires collagen synthesis and remodelling within an extra-cellular matrix scaffold.[37,39,40] Muller et al. found that during tendon healing the amount of collagen increases steadily and the repair callus reaches its largest size at around the fifth week.[37] However, healing of muscle tissue reaches peak proliferation at approximately week two to three.[41] These pathoanatomical differences between muscle and tendon should be considered during the rehabilitation process in combination with prognostic factors that may influence RTP.

Table 4
Best evidence synthesis of factors affecting RTP in intramuscular tendon tears *vs* no IMT involvement.

	Number of injuries	Studies with high risk of bias	Studies with low risk of bias	Association with RTP (↑/↓/~)	Best evidence synthesis	
					Association	Level of evidence
Severity (grade – BAMIC 1 *vs* 2, 1 *vs* 3, 2 *vs* 1, 2 *vs* 3, 3 *vs* 1, 3 *vs* 2)	308	Six studies[8,11,14,32,33,34]		↑ in four studies[8,11,14,32] ~ in two studies[33,34]	Unknown	Conflicting
Hamstring IMT involvement (*vs* no IMT involvement)	342	Five studies[7,8,11,14,34]	One study[5]	↑ in five studies[5,7,8,11,34] ~ in one study[14]	Yes	Moderate[a]
Quadriceps IMT involvement (*vs* no IMT involvement)	63	Two studies[12,33]		↑ in two studies[12,33]	Yes	Moderate
Craniocaudal length of ITM injury	140	One study[34]	One study[5]	~ in two studies[5,34]	No	Moderate
Cross-sectional area of ITM injury	134	Three studies[12,32,34]		↑ in three studies[12,32,34]	Yes	Moderate
Tendon waviness or loss of tendon tension	140	One study[34]	One study[5]	↑ in two studies[5,34]	Yes	Moderate

RTP = return to play, BAMIC = British Athletics Muscle Injury Classification, IMT = intramuscular tendon, ↓ indicates association with shorter time to RTP (negative association), ↑ indicates association with longer time to RTP (positive association), ~ indicates no association with time to RTP.
[a] Moderate evidence based on consistent findings in ≥75 % of studies.[28]

Table 5
Mean difference and 95 % CI for BAMIC 'b' *versus* 'c' injuries. Where sample sizes are small and/or standard deviations differ markedly, *P* values and CIs should be treated as approximate.

Author and date	Muscle group	Days to RTP; mean (SD)	Mean difference (95 % CI); *P*-value
McAleer et al. (2022)[33]	Quadriceps	2b = 23.5 (6.4); $n = 3$ 2c = 41.7 (18.7); $n = 4$ 3b = 34.0 (0.0); $n = 1$ 3c = 34.8 (20.6); $n = 4$	2c − 2b = 18.2 (−1.5, 37.9); $P = 0.105$[a] 3c − 3b = 0.8 (−72.5, 74.1); $P = 0.975$[a]
McAuley et al. (2022)[14]	Hamstring	2b = 26.0 (7.5); $n = 6$ 2c = 26.1 (8.2); $n = 8$	2c − 2b = 0.1 (−9.2, 9.4); $P = 0.982$
Pollock et al. (2022)[34]	Hamstring	2b = 20.8 (7.8);[b] $n = 11$ 2c = 34.5 (6.6);[b] $n = 7$	2c − 2b = 13.7 (7.0, 20.4); $P = 0.001$[*,a]
Tears et al. (2022)[32]	Hamstring	2b = 14.9 (10.3); $n = 18$ 2c = 23.8 (5.8); $n = 4$ 3b = 25.3 (21.0); $n = 3$ 3c = 23.5 (11.5); $n = 2$	2c − 2b = 8.9 (1.4, 16.3); $P = 0.032$[*,a] 3c − 3b = −1.8 (−30.4, 26.8); $P = 0.464$
Shamji et al. (2021)[8]	Hamstring	2b = 22.1 (9.5);[b] $n = 16$ 2c = 44.7 (30.7);[b] $n = 10$	2c − 2b = 22.6 (3.0, 42.2); $P = 0.024$[*,a]
Eggleston et al. (2020)[7]	Hamstring	2b = 24.3 (4.9); $n = 7$ 2c = 21.4 (5.0); $n = 5$ 3b = 19.5 (6.3); $n = 8$ 3c = 52.8 (30.1); $n = 10$	2c − 2b = −2.9 (−8.6, 2.8); $P = 0.182$ 3c − 3b = 33.3 (14.1, 52.5); $P = 0.005$[*,a]
Pollock et al. (2016)[11]	Hamstring	2b = 21.0 (10.2); $n = 17$ 2c = 27.0 (6.8); $n = 8$ 3b = 41.0 (0.0); $n = 1$ 3c = 84.0 (49.4); $n = 7$	2c − 2b = 6.0 (−0.8, 12.8); $P = 0.053$[a] 3c − 3b = 43.0 (−86.2, 172.2); $P = 0.447$[a]

SD = standard deviation, CI = confidence interval, RTP = return to play.
[*] $P \leq 0.05$.
[a] The direction of effect favoured the BAMIC 'b' classification for fewer days to RTP.
[b] Methods by Shi et al.[22] and Luo et al.[23] used to calculate the closest approximation of mean and SD from median and interquartile range, respectively.

4.3. BAMIC grading

This systematic review found moderate evidence for a positive association between time to RTP and BAMIC grading, supporting the use of this classification for muscle injury prognosis.[11,16] However, within this scale if there are any characteristics of a high-grade injury present then the injury is graded at the highest category. As such, the severity of intratendinous injuries is categorized by either CSA or longitudinal

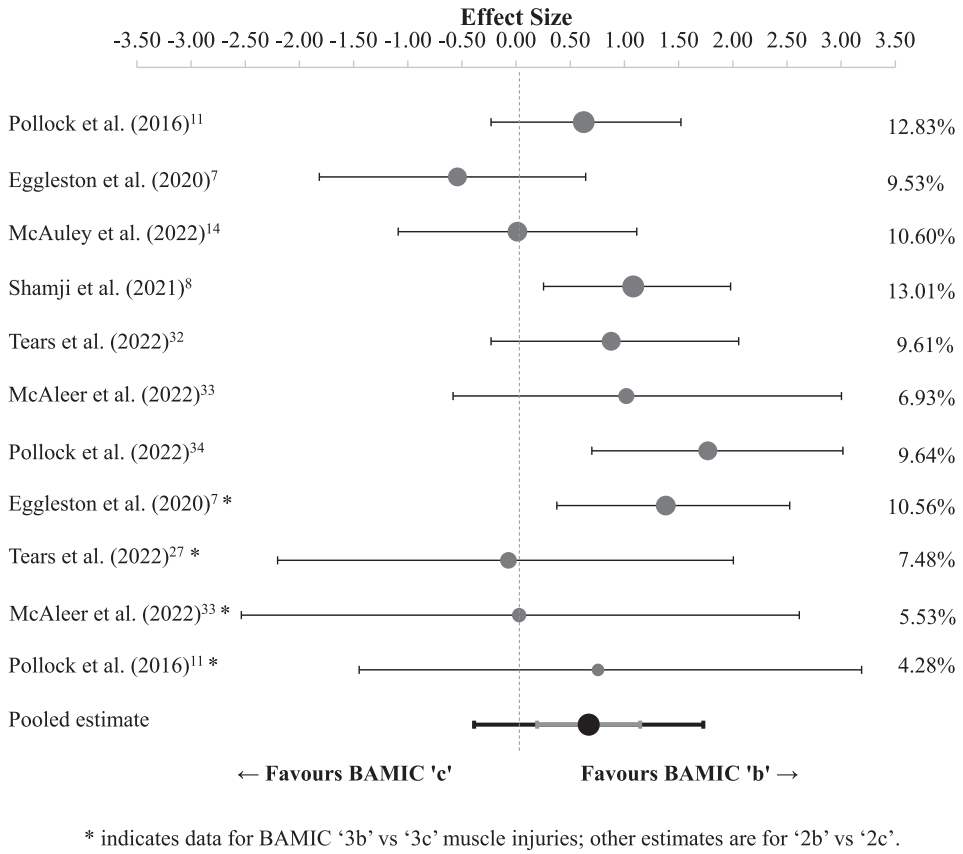

Fig. 2. Effect-size difference for grade two and three muscle injuries between BAMIC 'b' and BAMIC 'c' for time to return-to-play (160 injuries; see Table 5 for *n* values for individual estimates). Data are Hedges *g* statistic and 95 % confidence intervals. For the pooled estimate, the grey interval represents the 95 % confidence interval and the black interval represents the 95 % prediction interval. The % figures indicate each study's weighting in the calculation of the pooled estimate.

extent of tendon involvement. Intratendinous involvement can often extend for many centimetres, with longitudinal tears over 5 cm classified as BAMIC grade three.[15] The low threshold for longitudinal extent of tendon involvement for grade three injuries may explain the large variation in time to RTP noted within research (Table 2). The lack of association between longitudinal extent of IMT involvement and time to RTP could be explained by muscle–tendon unit integrity being maintained with longitudinal tears but disrupted with larger CSA involvement.[34] Van der Made et al. found that waviness was only present in IMT injuries with > 50 % disruption of tendon CSA and is consistent with a loss of longitudinal-plane tendon tension and tensile strength.[5] In longitudinal split tears, the orientation of the tear is aligned to the parallel fibre orientation of the tendon and although there is some failure of the myofibrillar attachments, analogous to the failure of struts along a suspension bridge, the IMT can maintain its tendon tension.[6]

The orientation of IMT involvement may be a greater prognostic factor than initially considered. McAuley et al. found no significant difference in time to RTP between BAMIC grade '2b' and '2c' injuries in professional footballers. From the nine injuries they examined with IMT involvement, eight were longitudinal in orientation and one was reported as transverse (CSA involvement).[14] The BAMIC scale could be modified to accommodate the length of tendon injury as a differentiator between the categorization of grade '2c' and '3c' injuries, to take into consideration the integrity, or failure, of the muscle–tendon unit and consequently time to RTP.

4.4. Limitations and future recommendations

The quality of a systematic review is dependent on the quality of the included studies. This review pooled data from studies with a high risk of bias and moderate heterogeneity; the pooled effect should therefore be interpreted with caution. The CIs around both the effect size and the overall mean difference in days to RTP are fairly wide, indicating some uncertainty in these estimates.

In addition to comparing '2b' *vs* '3b' injuries with '2c' *vs* '3c' injuries in the meta-analysis we also produced separate subgroup estimates for '2b' *vs* '2c' and for '3b' *vs* '3c' injuries; however, owing to sparse data the effect size for '3b' *vs* '3c' injuries should be interpreted cautiously. Additionally, although the review contains a large total number of injuries, the number of injuries in specific severity categories is sometimes quite small.

To take account of study heterogeneity, a random-effects meta-analysis was performed, which rests on the assumption that the included studies estimate a distribution of underlying population effects. However, had a fixed-effect analysis been performed, which assumes that all included studies are estimating a single underlying population effect, a very similar pooled effect-size would have been estimated: 0.69 (95 % CI 0.32, 1.06), $P < 0.001$. Language bias is a possible limitation of this review as only English language studies were searched.

Given the identified limitations of this review and the methodological limitations of included studies discussed above, future research should use standardized criteria for RTP and blinding of clinicians to prognostic features on MRI. Lastly, adjusting of the BAMIC scale could be considered for longitudinal tears of the IMT.

5. Conclusion

Due to the high risk of bias and heterogeneity across studies, providing an accurate RTP prognosis for lower-limb muscle injuries involving the IMT remains difficult. This review found no strong evidence for a difference in time to RTP between injuries with and without IMT involvement. Pooled analysis demonstrated a medium effect-size difference between BAMIC 'b' *vs* 'c' injuries, with a more favourable prognosis for '2b' muscle injuries over intratendinous '2c' injuries. The percentage CSA of tendon disruption, presence of waviness and loss of tendon tension on MRI all showed moderate evidence for an association with increased time to RTP, but the longitudinal extent of tendon disruption showed no association. Further high-quality research is required to provide strong evidence for an accurate prognosis of muscle injuries with IMT involvement.

Funding information

No external financial support.

Confirmation of ethical compliance

This research was approved by Keele University Ethics Committee.

CRediT authorship contribution statement

Catherine Beattie: Conceptualization, writing — original draft, methodology, data curation, project administration, writing — review and editing. **Richard Barnett**: Supervision, writing — review and editing. **Jennifer Williams**: Methodology. **Julius Sim**: Methodology, analysis, data curation, writing — review and editing. **Samuel Pullinger**: Writing — review and editing, methodology.

Declaration of interest statement

None declared.

Acknowledgements

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jsams.2023.10.002.

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